

Rahul Rawat

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Fraunhofer SIT

Hiring Team

Application: Werkstudierende in the NLP research area at Fraunhofer SIT

Dear Hiring Team,

I am applying for the Werkstudierende role in the NLP research area at Fraunhofer SIT in Darmstadt, reference 81228. The brief on implementing NLP approaches around feature extraction, style change detection, AI generated text detection, and authorship attribution, plus applying machine learning methods spanning zero and few shot learning, fine tuning of foundation models, Transformer architectures, CNNs, and classical baselines, is exactly the shape of NLP work I have been building this year.

My Hybrid RAG Orchestrator is a working NLP system with a custom decision making router over Llama 3.1 8b via Groq and LangChain, a ChromaDB vector store with local persistence, and HuggingFace MiniLM L6 v2 embeddings for semantic retrieval. Building it exercised the LLM based feature extraction, embedding, and prompt engineering side of the role and taught me the practical side of Transformer inference in a production style pipeline. The Streamlit interface on top gave me hands on experience shipping user facing NLP tools, in line with your interest in Streamlit, Gradio, and similar web UIs.

In CreditIQ I ran rigorous machine learning evaluation across classifiers, using SHAP for driven subgroup analysis and computing standard metrics including ROC AUC and confusion matrices. In my Diabetes Prediction Bachelor Thesis I compared six classifiers on a clinical dataset with 10 fold cross validation and chose ROC AUC as the honest headline metric for an imbalanced target. That evaluation discipline transfers directly to the authorship attribution and verification benchmarks in your work.

I am proficient in Python, comfortable with Transformer style architectures, CNNs, and classical baselines, and hold the NVIDIA Building LLM Applications With Prompt Engineering certificate. I would be glad to work 40 to 80 hours per month around my studies and to discuss how the role could extend into a Master Thesis at Fraunhofer SIT.

Kind regards,

Rahul Rawat